Results for algebra2005

Current results grouped by recommendation-graph parameters | Snapshot: 2026-10-06 15:25 CST

Each group uses the same six model variants, N = 5, 10, ..., 50, and four Ep response thresholds.

Bold = best; underline = second best by unrounded value. Rankings require all six models.

Parameter group	Completed models	Run status
1. theta_m=0.8, theta_f=0.5, rec=50	**6/6**	completed
2. theta_m=0.8, theta_f=0.4, rec=50	**6/6**	completed
3. theta_m=0.8, theta_f=0.8, rec=10	**6/6**	completed

Metric source

Data_Fin_New/_sensitivity_runs/<variant>/<dataset>/sensitivity_seed2024/
SemanticConvE_<model>/seed2024/eval/metrics.json

Group 1: theta_m=0.8, theta_f=0.5, rec=50

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.750**	0.220	0.699	0.721	0.735	0.717
10	**0.743**	0.219	0.699	0.706	0.732	0.695
15	0.731	0.217	0.699	0.695	**0.732**	0.682
20	0.728	0.217	0.699	0.691	**0.728**	0.670
25	**0.726**	0.230	0.699	0.687	0.723	0.661
30	**0.723**	0.237	0.706	0.681	0.721	0.657
35	**0.718**	0.237	0.709	0.674	0.718	0.649
40	**0.713**	0.242	0.711	0.668	0.712	0.644
45	0.705	0.248	**0.716**	0.663	0.707	0.638
50	0.699	0.253	**0.715**	0.656	0.701	0.632

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.992**	0.964	0.991	0.989	0.991	0.990
10	**0.991**	0.961	0.991	0.988	0.990	0.989
15	**0.991**	0.960	0.991	0.988	0.991	0.990
20	**0.991**	0.962	0.991	0.988	0.990	0.990
25	0.990	0.964	**0.991**	0.989	0.989	0.991
30	0.990	0.965	0.991	0.989	0.989	**0.991**
35	0.989	0.966	0.991	0.988	0.988	**0.991**
40	0.989	0.966	0.991	0.988	0.988	**0.991**
45	0.989	0.966	**0.991**	0.988	0.988	0.990
50	0.988	0.967	**0.990**	0.988	0.988	0.990

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 1: theta_m=0.8, theta_f=0.5, rec=50

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.092	0.018	**0.165**	0.004	-0.019	-0.012
10	-0.031	0.037	**0.237**	0.086	0.062	0.040
15	0.087	0.017	**0.251**	0.116	0.013	0.063
20	0.031	0.049	**0.263**	0.137	0.100	0.088
25	0.101	0.067	**0.245**	0.120	0.153	0.128
30	**0.163**	0.071	-0.030	0.133	0.124	0.139
35	0.170	0.055	0.053	0.117	**0.178**	0.127
40	**0.158**	0.086	0.125	0.104	0.132	0.132
45	0.019	0.064	-0.079	0.123	0.131	**0.141**
50	0.017	0.092	0.078	0.122	0.041	**0.143**

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.101	0.018	**0.162**	-0.002	-0.049	-0.023
10	-0.044	0.037	**0.235**	0.075	0.027	0.014
15	0.073	0.017	**0.250**	0.108	0.005	0.041
20	0.024	0.047	**0.262**	0.129	0.095	0.065
25	0.084	0.063	**0.245**	0.109	0.152	0.097
30	**0.160**	0.066	-0.030	0.121	0.123	0.120
35	0.169	0.036	0.053	0.104	**0.178**	0.105
40	**0.158**	0.073	0.125	0.080	0.132	0.108
45	0.019	0.052	-0.079	0.095	**0.131**	0.119
50	0.017	0.080	0.078	0.093	0.041	**0.124**

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 1: theta_m=0.8, theta_f=0.5, rec=50

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.252	0.007	**0.108**	-0.081	-0.197	-0.120
10	-0.269	0.022	**0.179**	-0.038	-0.249	-0.122
15	-0.238	-0.018	**0.203**	-0.022	-0.288	-0.110
20	-0.322	-0.011	**0.218**	-0.003	-0.220	-0.093
25	-0.244	0.033	**0.201**	-0.025	-0.097	-0.060
30	-0.036	**0.028**	-0.179	-0.029	-0.077	-0.046
35	-0.031	-0.013	-0.270	-0.053	**0.056**	-0.064
40	0.010	**0.042**	-0.278	-0.099	0.025	-0.055
45	-0.101	0.014	-0.465	-0.095	**0.054**	-0.044
50	-0.166	**0.040**	-0.377	-0.087	-0.069	-0.035

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.348	-0.089	**-0.012**	-0.168	-0.304	-0.213
10	-0.373	-0.085	**0.007**	-0.179	-0.344	-0.254
15	-0.339	-0.132	**0.001**	-0.175	-0.380	-0.281
20	-0.403	-0.146	**0.000**	-0.180	-0.324	-0.278
25	-0.338	-0.126	**-0.029**	-0.214	-0.260	-0.264
30	-0.206	**-0.105**	-0.404	-0.221	-0.307	-0.264
35	-0.189	**-0.129**	-0.383	-0.247	-0.235	-0.271
40	-0.198	**-0.111**	-0.326	-0.277	-0.302	-0.267
45	-0.320	**-0.132**	-0.478	-0.260	-0.267	-0.247
50	-0.347	**-0.098**	-0.383	-0.263	-0.351	-0.251

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: theta_m=0.8, theta_f=0.4, rec=50

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.699	0.201	0.699	**0.713**	0.699	0.677
10	0.699	0.202	0.699	0.707	**0.708**	0.678
15	**0.715**	0.202	0.699	0.705	0.714	0.679
20	**0.722**	0.202	0.699	0.698	0.719	0.681
25	0.720	0.203	0.699	0.695	**0.721**	0.681
30	0.720	0.204	0.701	0.690	**0.723**	0.681
35	0.718	0.205	0.709	0.686	**0.720**	0.678
40	0.716	0.206	0.711	0.679	**0.717**	0.675
45	0.714	0.206	0.714	0.670	**0.714**	0.670
50	0.710	0.207	**0.715**	0.660	0.710	0.665

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.991	0.953	0.991	**0.991**	0.991	0.991
10	0.991	0.950	0.991	0.991	**0.992**	0.988
15	0.992	0.949	0.991	0.989	**0.992**	0.988
20	**0.991**	0.951	0.991	0.989	0.991	0.989
25	0.991	0.952	**0.991**	0.989	0.990	0.989
30	**0.991**	0.953	0.990	0.988	0.991	0.990
35	0.990	0.955	**0.991**	0.987	0.991	0.990
40	0.990	0.956	**0.991**	0.988	0.990	0.990
45	0.989	0.957	**0.991**	0.988	0.989	0.989
50	0.989	0.958	**0.990**	0.989	0.989	0.989

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: theta_m=0.8, theta_f=0.4, rec=50

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.172**	-0.042	0.165	0.030	0.166	0.084
10	0.229	0.031	**0.231**	0.096	0.122	0.112
15	-0.022	0.013	**0.245**	0.091	0.042	0.129
20	-0.057	0.018	**0.257**	0.084	-0.021	0.125
25	0.117	0.050	**0.248**	0.099	0.031	0.114
30	0.125	0.084	**0.168**	0.098	0.035	0.097
35	0.109	0.077	-0.002	0.118	**0.183**	0.107
40	0.144	0.114	0.115	0.101	**0.194**	0.119
45	0.192	0.115	0.031	0.113	**0.246**	0.129
50	0.173	0.128	0.022	0.112	**0.188**	0.123

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.167**	-0.042	0.162	0.022	0.163	0.075
10	0.227	0.031	**0.230**	0.082	0.120	0.100
15	-0.023	0.012	**0.244**	0.077	0.042	0.114
20	-0.098	0.018	**0.256**	0.069	-0.042	0.110
25	0.105	0.048	**0.248**	0.073	0.008	0.098
30	0.124	0.083	**0.168**	0.070	0.029	0.077
35	0.109	0.076	-0.115	0.091	**0.182**	0.087
40	0.144	0.113	0.089	0.072	**0.193**	0.094
45	0.192	0.115	0.026	0.076	**0.246**	0.105
50	0.173	0.127	0.021	0.077	**0.188**	0.097

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: theta_m=0.8, theta_f=0.4, rec=50

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.105	-0.048	**0.106**	-0.067	0.103	0.014
10	0.166	0.029	**0.168**	-0.057	0.057	0.021
15	-0.085	0.008	**0.197**	-0.088	-0.067	0.007
20	-0.306	-0.009	**0.221**	-0.097	-0.286	0.002
25	-0.266	0.036	**0.215**	-0.083	-0.308	-0.022
30	-0.198	0.076	**0.134**	-0.115	-0.333	-0.071
35	-0.062	**0.054**	-0.265	-0.096	-0.121	-0.080
40	0.053	**0.104**	-0.304	-0.109	0.004	-0.084
45	0.137	0.109	-0.398	-0.102	**0.160**	-0.078
50	0.122	0.120	-0.427	-0.113	**0.163**	-0.087

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.016	-0.158	**-0.009**	-0.175	-0.012	-0.087
10	-0.003	-0.113	**0.011**	-0.182	-0.138	-0.138
15	-0.310	-0.101	**0.004**	-0.220	-0.315	-0.181
20	-0.427	-0.104	**0.004**	-0.238	-0.434	-0.199
25	-0.345	-0.096	**-0.016**	-0.246	-0.401	-0.238
30	-0.320	**-0.036**	-0.200	-0.273	-0.380	-0.281
35	-0.274	**-0.034**	-0.407	-0.259	-0.239	-0.279
40	-0.261	**0.017**	-0.348	-0.259	-0.231	-0.277
45	-0.211	**0.039**	-0.409	-0.247	-0.155	-0.281
50	-0.193	**0.060**	-0.430	-0.258	-0.189	-0.293

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: theta_m=0.8, theta_f=0.8, rec=10

6/6 models completed | completed | ranked across six models

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.748	0.682	**0.788**	0.693	0.748	0.744
10	0.735	0.650	**0.766**	0.668	0.728	0.729
15	0.734	0.621	**0.744**	0.644	0.729	0.714
20	**0.726**	0.588	0.718	0.626	**0.726**	0.693
25	0.697	0.544	**0.703**	0.608	0.701	0.668
30	0.683	0.521	**0.701**	0.588	0.672	0.649
35	0.652	0.504	**0.673**	0.571	0.652	0.625
40	**0.645**	0.488	0.644	0.557	0.629	0.603
45	**0.622**	0.464	0.616	0.541	0.621	0.592
50	0.600	0.444	**0.601**	0.526	0.597	0.578

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.989	0.982	**0.994**	0.990	0.989	0.986
10	0.988	0.983	**0.992**	0.988	0.987	0.988
15	0.988	0.982	**0.990**	0.989	0.987	0.986
20	0.986	0.981	0.987	**0.988**	0.986	0.985
25	0.986	0.980	**0.988**	0.987	0.985	0.985
30	0.986	0.980	**0.986**	0.986	0.986	0.986
35	0.986	0.980	**0.988**	0.985	0.986	0.986
40	0.986	0.979	**0.988**	0.985	0.987	0.986
45	0.987	0.978	**0.987**	0.985	0.987	0.986
50	0.986	0.977	**0.988**	0.984	0.987	0.986

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: theta_m=0.8, theta_f=0.8, rec=10

6/6 models completed | completed | ranked across six models

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.152	**0.040**	-0.104	-0.009	-0.026	-0.018
10	**0.145**	0.098	-0.057	0.060	0.053	0.021
15	0.073	0.111	**0.192**	0.072	-0.009	0.072
20	0.098	0.166	**0.307**	0.101	0.004	0.128
25	0.269	0.181	0.299	0.141	**0.304**	0.185
30	**0.338**	0.163	0.225	0.131	0.283	0.250
35	0.316	0.086	**0.399**	0.144	0.306	0.280
40	0.241	0.068	**0.330**	0.150	0.151	0.208
45	0.120	0.044	0.089	0.139	**0.198**	0.139
50	0.293	0.072	-0.007	0.128	**0.399**	0.178

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.235	**0.010**	-0.110	-0.034	-0.033	-0.029
10	**0.140**	0.079	-0.153	0.028	0.052	0.000
15	0.073	0.106	**0.174**	0.052	-0.009	0.052
20	0.098	0.165	**0.306**	0.084	-0.125	0.111
25	0.185	0.179	**0.299**	0.127	0.247	0.173
30	**0.318**	0.153	0.225	0.120	0.263	0.239
35	0.310	0.058	**0.399**	0.133	0.301	0.275
40	0.237	0.044	**0.330**	0.138	-0.004	0.202
45	0.039	0.024	0.089	0.121	**0.150**	0.121
50	0.277	0.057	-0.007	0.108	**0.389**	0.161

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: theta_m=0.8, theta_f=0.8, rec=10

6/6 models completed | completed | ranked across six models

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.347	-0.104	-0.313	-0.138	-0.104	**-0.093**
10	**0.020**	-0.053	-0.410	-0.110	-0.036	-0.090
15	-0.042	**0.003**	-0.206	-0.084	-0.036	-0.048
20	-0.013	0.144	**0.268**	-0.032	-0.311	-0.038
25	0.062	0.170	**0.221**	0.021	0.038	0.030
30	**0.283**	0.141	0.182	0.016	0.122	0.170
35	**0.296**	0.041	0.214	0.044	0.274	0.234
40	0.132	0.009	**0.299**	0.053	-0.021	0.165
45	0.020	-0.003	-0.016	0.034	**0.139**	0.078
50	0.271	0.025	-0.040	0.021	**0.384**	0.100

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.428	-0.202	-0.409	-0.246	-0.286	**-0.202**
10	-0.206	**-0.182**	-0.436	-0.251	-0.300	-0.282
15	-0.317	**-0.190**	-0.231	-0.254	-0.368	-0.265
20	-0.307	-0.022	**0.057**	-0.229	-0.407	-0.233
25	-0.164	**0.043**	-0.060	-0.194	-0.004	-0.200
30	**0.211**	0.079	-0.006	-0.208	-0.084	-0.091
35	0.055	-0.020	**0.178**	-0.191	-0.008	0.020
40	-0.131	-0.084	**0.130**	-0.170	-0.239	-0.049
45	-0.302	**-0.103**	-0.344	-0.184	-0.257	-0.203
50	0.137	-0.083	-0.432	-0.197	**0.279**	-0.195

Seed 2024; single-run values. No across-seed standard deviation is available.